Dividing fractions

20 questions, including 2 challenges. Write your work and answers on separate paper.

1. Write $\frac{1}{2}$ as a decimal and a percent.

2. Write $\frac{1}{4}$ as a decimal and a percent.

3. Write $\frac{3}{4}$ as a decimal and a percent.

4. Write $\frac{1}{8}$ as a decimal and a percent.

5. Find 140% of 30 points.

6. A number line uses quarters from 0 to 2. Where does $\frac{7}{4}$ belong?

7. $96 \div 8$

8. Find $3\ \frac{1}{2} \div \frac{3}{4}$.

9. Which is the reciprocal of $2\ \frac{1}{2}$: $\frac{2}{5}$ or $\frac{5}{2}$? Explain using a product of 1.

10. Find $1\ \frac{1}{2} \div \frac{3}{4}$.

11. $\frac{4}{3} \div \frac{3}{5}$

12. $\frac{4}{3} \times \frac{4}{5}$

13. Zoe writes $\frac{7}{8} \div \frac{1}{4} = \frac{8}{7} \times \frac{1}{4} = \frac{2}{7}$. Explain which number should be inverted.

14. How many $\frac{3}{5}$-sized groups fit in $\frac{9}{10}$? Include a part of a group.

15. How many $\frac{5}{6}$-cup scoops fit in 2 cups? A partial scoop is allowed.

16. Share $\frac{3}{8}$ liter equally among 6 bottles. Find each share.

17. Find $3\ \frac{1}{6} \div \frac{4}{5}$. Check by multiplication.

18. A rectangle has area $\frac{2}{5}$ square meter and width $\frac{2}{5}$ meter. Find its length.

CHALLENGE / Explain your plan and check the result

C1. There are $3\ \frac{1}{2}$ kilograms of rice. Each meal uses $\frac{2}{5}$ kilogram. Find the greatest number of complete meals and the rice left over.

C2. A poster has area $1\ \frac{7}{8}$ square meters and width $\frac{3}{4}$ meter. Another poster has the same area and width $\frac{5}{8}$ meter. Find both lengths. Which must be longer, and why?